

Enhancing Real-Time Lip-Sync and Avatar Quality for Online Meetings

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Abstract—In recent years, the use of avatars in real-time online meetings has garnered significant attention, yet ensuring accurate and visually appealing lip synchronization remains a challenge, especially under varying lighting conditions and with the need for real-time performance. In this work, we address the problems of blurry lip motion and visual degradation often introduced by state-of-the-art audio-driven lip-sync models such as MuseTalk. To resolve these issues, we propose a novel pipeline that combines the MuseTalk model for initial lip-sync generation with the CodeFormer model to enhance image quality and correct blurry lips and motion artifacts. Furthermore, we introduce a novel image-processing algorithm designed to dynamically compensate for irregular and low-lighting scenarios, thereby improving the realism and robustness of avatar representation. Our end-to-end system is optimized for real-time performance, making it suitable for live online meetings. Experimental results demonstrate that our approach produces more natural and visually stable avatars, significantly outperforming baseline methods in both subjective and objective evaluations. This work paves the way for seamless, real-time avatar-driven communication in diverse lighting environments. Github link for the code base is given in this [link](#).

I. INTRODUCTION

In recent years, the generation of photorealistic human videos[1, 2] has gained significant attention, particularly in applications that require realistic facial expressions and accurate lip synchronization. One of the most compelling challenges in this domain is the generation of human face videos that precisely sync lip movements with a given audio stream. This task holds immense potential for applications in virtual avatars, film dubbing, telepresence, and accessibility tools for the hearing impaired.

Approaches to audio-driven human video generation can be broadly classified into two categories. The first involves image-to-video [3] generation, where a single still image of a person and an accompanying audio stream are used to generate a full video. These methods not only produce lip-synced speech but also attempt to autonomously synthesize naturalistic head and eye movements in accordance with the audio cues. Despite achieving impressive visual fidelity, their dependence on iterative denoising processes—especially in diffusion-based frameworks—significantly limits their feasibility for real-time or resource-constrained applications. The second line of work explores one-shot video dubbing, a video-to-video (V2V) paradigm that seeks to preserve the original head and eye dynamics of the input video while

selectively modifying only the lip region to match new speech. Unlike image-to-video methods, these techniques do not require retraining for each subject or scene, making them more adaptable in practical scenarios. One-shot dubbing approaches are primarily built upon either diffusion-based models [4] or GAN-based architectures [5]. Although diffusion models offer high-quality outputs and robust lip synchronization, their computational demands—both in terms of training and multi-step inference—pose challenges for efficient deployment.

GAN-based methods, while faster due to single-step inference, often suffer from low video quality, producing blurry or distorted facial regions and failing to preserve the actor's identity. These issues limit their practical use, especially in applications demanding high visual fidelity. Despite known training instabilities, their real-time generation capability keeps them relevant for time-sensitive scenarios.

To address the limitations of existing methods, we build upon MuseTalk [6], a GAN-based real-time one-shot video-dubbing framework. MuseTalk mitigates key training challenges in GAN-based dubbing using a two-stage training pipeline and a spatio-temporal sampling strategy, ensuring identity consistency and accurate lip synchronization. While it achieves high-fidelity video generation via video-to-video dubbing, it often compromises quality in the lower facial regions due to its limited focus on audiovisual consistency. Additionally, prior works have observed that high-resolution inputs can lead to degraded outputs, as the increased detail introduces greater nonlinearity, often resulting in higher training loss and reduced model stability. To overcome these challenges, we propose a two-staged network that results in both high fidelity and better quality videos. This is done using the GAN-based muse talk model cascaded with a transformer based prediction network, named CodeFormer [7], to model the global composition and context of the low-quality faces for code prediction, enabling generation of natural faces that closely approximate the target faces.

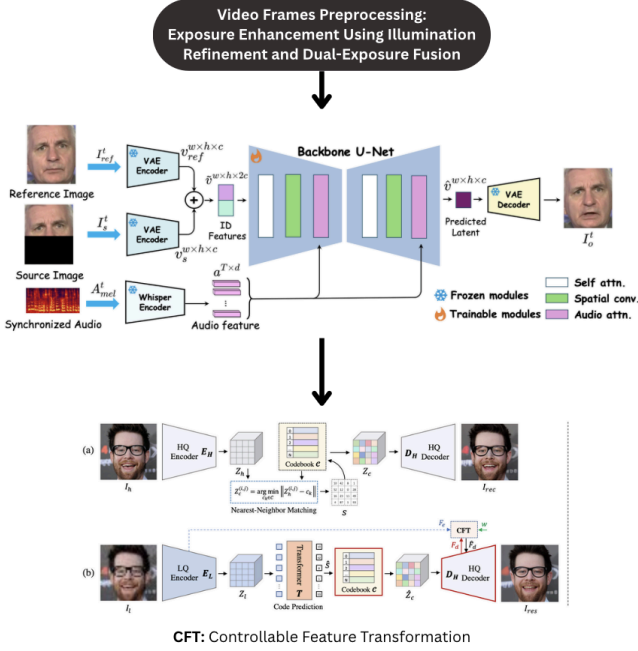
As an additional enhancement, we incorporate a preprocessing step to improve the quality of input videos, particularly those suffering from low exposure or poor lighting conditions. To this end, we adopt a Robust Exposure Correction technique [8, 9] which effectively balances under- and over-exposed regions to produce well-illuminated frames. By correcting exposure before feeding the video into the MuseTalk pipeline, we ensure that

the model receives clear and visually consistent inputs. This preprocessing step significantly aids in achieving more accurate lip synchronization and overall visual quality, especially in challenging lighting scenarios.

II. LIP-SYNCING AVATAR PIPELINE

This section discusses our overall working pipeline of our proposed architecture.

A. Overview



The proposed method consists of three main components:

1. **Image Pre-processing:** This stage addresses issues related to low or uneven lighting in videos and images. It enhances visual quality to ensure better performance in subsequent deep learning tasks. It is primarily based on Retinex theory and has been optimized for high-speed performance, making it suitable for real-time applications.
2. **Lip-Sync Generation using MuseTalk:** MuseTalk is a state-of-the-art audio-driven lip-sync model capable of generating realistic talking head videos from audio inputs. In our pipeline, MuseTalk takes the enhanced video frames and the corresponding audio as input to generate temporally consistent and expressive lip movements that synchronize accurately with speech, serving as the foundation for avatar animation in real-time communication scenarios.
3. **Image Enhancement using CodeFormer:** CodeFormer is a robust face restoration model designed to reconstruct high-quality facial details from low-resolution or distorted inputs. In our pipeline, CodeFormer is applied to the frames generated by MuseTalk to correct visual artifacts such as blurry lips, motion distortions, and inconsistent facial textures.

Together, these components form an efficient pipeline for generating high-quality, lip-synced avatar videos, even under challenging visual conditions.

B. Image Preprocessing

Exposure Enhancement Using Illumination Refinement and Dual-Exposure Fusion

To address both underexposed and overexposed regions in real-time video feeds, we adopt a combination of two complementary algorithms:

- **LIME:** Low-Light Image Enhancement via Illumination Map Estimation
- **DUAL:** Dual-Exposure Fusion for Low-Light Image Enhancement

These methods are based on illumination map estimation, structure-aware refinement, and exposure fusion principles.

Initial Illumination Estimation

Given an RGB image $\mathbf{I}$ of size $H \times W \times 3$, we compute the initial illumination map $\hat{\mathbf{L}}$ by selecting the maximum intensity among the three channels at each pixel:

$$\hat{\mathbf{L}}(x, y) = \max \{ \mathbf{I}_R(x, y), \mathbf{I}_G(x, y), \mathbf{I}_B(x, y) \}$$

This rough map indicates the brightest component and serves as a baseline for enhancement.

Illumination Map Refinement

To improve the smoothness and preserve edges in the illumination map, we refine it using a structure-aware optimization technique:

1. Spatial Gaussian Kernel

A kernel $\mathbf{K}(\mathbf{i}, \mathbf{j})$ is defined as:

$$\mathbf{K}(\mathbf{i}, \mathbf{j}) = \exp [-\|\mathbf{i}, \mathbf{j}\| - (s/2, s/2)^2 / (2\sigma^2)]$$

This gives higher weight to closer neighbors in a spatial window of size $s \times s$.

2. Directional Weights

Directional edge-aware weights $\mathbf{w}_x$ and $\mathbf{w}_y$ are calculated as:

$$\begin{aligned} \mathbf{w}_x &= (\mathbf{K} * 1) / (|\mathbf{K} * \partial_x \hat{\mathbf{L}}| + \epsilon) \\ \mathbf{w}_y &= (\mathbf{K} * 1) / (|\mathbf{K} * \partial_y \hat{\mathbf{L}}| + \epsilon) \end{aligned}$$

Where:

- $\partial_x \hat{\mathbf{L}}$ and $\partial_y \hat{\mathbf{L}}$ are the image gradients in X and Y directions
- ϵ is a small constant to avoid division by zero

3. Structure-Aware Optimization

We then minimize the following cost function:

$$\min \|L - \hat{L}\|^2 + \lambda \sum [w_x(x, y)(\partial_x L)^2 + w_y(x, y)(\partial_y L)^2]$$

This balances fidelity to the initial illumination with smoothness guided by edges. The final solution is computed by solving the sparse linear system:

$$(I + \lambda F) \cdot L = \hat{L}$$

Where F is a Laplacian matrix built from w_x and w_y .

Dual-Exposure Fusion

DUAL improves both bright and dark areas by fusing multiple enhanced versions:

- I_u : Underexposed-enhanced image (from I)
- I_o : Overexposed-enhanced image (by applying the same algorithm on inverted image $1 - I$, then inverting back)
- I : Original image

These are fused using the **Mertens Exposure Fusion** technique based on:

- Local Contrast (C)
- Color Saturation (S)
- Well-Exposedness (E)

Fusion Weights:

$$W_k = C_k^b \cdot S_k^b \cdot E_k^b$$

($k \in \{I, I_u, I_o\}$)

Final Output:

$$I_f = [\sum W_k \cdot I_k] / [\sum W_k]$$

Typical values:

- $b_c = 1$ (contrast),
- $b_s = 1$ (saturation),
- $b_e = 1$ (well-exposedness)

Post-Processing

The fused image I_f is scaled to the $[0, 255]$ range and optionally passed through denoising or sharpening filters to improve final appearance.

C. *MuseTalk*

Framework

1. Latent space encoding with VAE

Use a pretrained VAE to encode occluded lower face image I_s^t and facial reference image I_{ref}^t , and output latent feature

$v^{w \times h \times 2c}$ by concatenating $v_{ref}^{w \times h \times c}$ and $v_m^{w \times h \times c}$ along channel dimension
 w : width of feature
 h : height of feature
 c : channel

2. Audio Preprocessing

Pretrained whisper encoder to extract features from sequence input audio segment with length T and centered at time t . This segment is resampled at 16000 Hz and transformed to 80-channel log magnitude Mel spectrogram. the input audio feature is $A_d^t \in R^{T \times 80}$ and the output audio feature is $a^{T \times d}$
 d : dimension of audio feature

3. Multi scale fusion U-Net

Aligns two heterogeneous modalities, audio and visual by fusing audio and visual features across various scales and modelling conditional distributions of the form $p(\hat{v}|a)$. It generates the final output I_o^t .

Loss functions

1. Reconstruction loss

L1 loss for reconstruction to minimize pixel-wise differences and ensure colour and structural consistency.

$$L_{rec} = \|I_o^t - I_{gt}^t\|_1$$

2. Perception loss

Emphasizes perceptual similarity by comparing high level features, enhancing details and visual realism.

$$L_p = \|\nu(I_o^t) - \nu(I_{gt}^t)\|_2$$

3. GAN loss

Encourages model to learn subtle details by challenging a discriminator.

$$L_G = E \left[\log \left(1 - D(I_o^t) \right) \right]$$

$$L_D = E \left[\log \left(D(I_o^t) \right) \right]$$

$$L_{GAN} = L_G + L_D$$

4. Lip sync loss

Improve synchronization of lip movements in dubbed videos

$$L_{sync} = \frac{1}{N} \sum_i -\log \left[P_{sync} \left(SyncNet \left(A_{mel}^i, I_o^i \right) \right) \right]$$

5. Total loss

$$L = L_{rec} + \lambda L_p + \mu L_{GAN} + \phi L_{sync}$$

D. *CodeFarmer*

CodeFormer addresses the challenge of restoring high-quality facial details by operating in a discrete representation space, which reduces the uncertainty of the restoration process and enhances degraded inputs with

realistic textures. To compensate for the loss of local details commonly present in low-quality inputs, CodeFormer employs a transformer module that captures the global composition of natural faces, effectively recovering the missing information and producing visually coherent outputs. It first leverages vector quantization by pre-training a quantized autoencoder through self-reconstruction, resulting in a discrete codebook and decoder. This codebook acts as a prior, guiding the restoration process. The transformer then predicts accurate code combinations from degraded inputs, enabling natural face synthesis.

MuseTalk achieves high-fidelity lip-synced video generation but it often struggles with visual realism, particularly in the mouth region. To overcome this limitation, we cascade MuseTalk with CodeFormer, combining the GAN-based video dubbing and transformer-based facial detail restoration. This integrated approach leads to enhanced visual quality and more realistic outputs, especially in challenging facial regions.

The training of this method is divided into three stages accordingly:

- **Codebook Learning (Stage I)** : Instead of directly mapping LQ-HQ images, first the quantized autoencoder is pre-trained to learn a context-rich codebook, which improves network expressiveness as well as robustness against degradation.
- **Codebook Lookup Transformer Learning (Stage II)**: Due to corruptions of textures in LQ faces, the nearest-neighbour (NN) matching usually fails in finding accurate codes for face restoration. To mitigate this problem, a Transformer block is employed to model global interrelations for better code prediction.
- **Controllable Feature Transformation (CFT) (Stage III)**: The CFT is leveraged to investigate the tradeoff between high fidelity and high quality. This is applied by controlling the information flow from LQ encoder to HQ decoder.

III. RESULTS

To evaluate the effectiveness of our proposed pipeline for real-time avatar-based video communication under challenging lighting conditions, we conducted both qualitative and quantitative experiments. The results presented in this section demonstrate the impact of the exposure enhancement preprocessing, the fidelity of lip synchronization via MuseTalk, and the visual improvements achieved through CodeFormer-based post-processing. Additionally, we analyze the trade-offs in processing time with and without disk-based image buffering, offering insights into the feasibility of real-time deployment.

In scenarios where input video frames exhibit irregular or

poor lighting conditions, particularly low-light environments, we incorporate a preprocessing step using a state-of-the-art low-light image enhancement algorithm developed by Nie et al. [^1]. This algorithm is implemented from the open-source repository [Low-light-Image-Enhancement](#), which leverages a **zero-reference learning approach** to improve visibility in dark images while preserving color consistency and structural integrity.

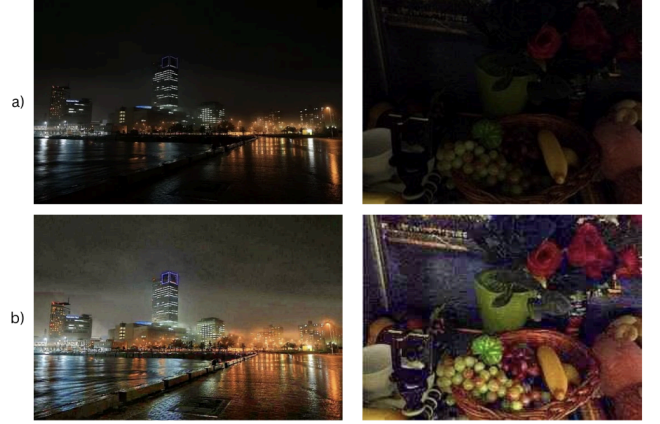
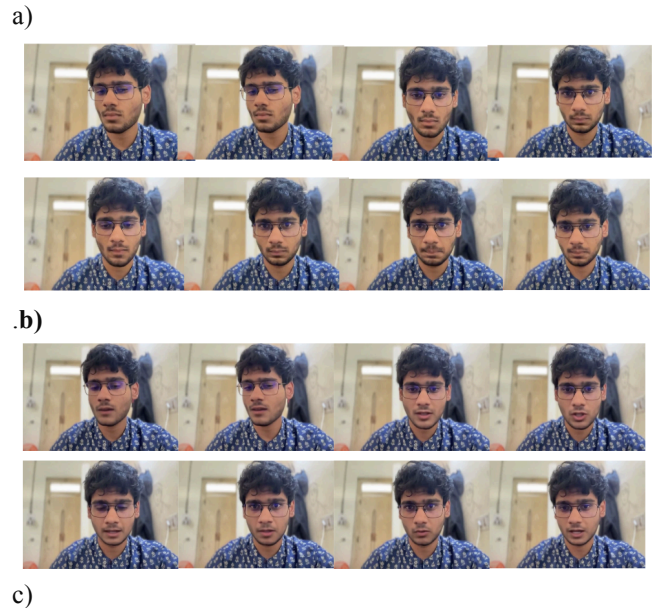


Fig. 2. a) Raw input images captured under low-light conditions. b) Corresponding enhanced outputs showing significant improvements in brightness, contrast, and visibility, while preserving structural details and minimizing artifacts.

We utilized the Zero-Reference Deep Curve Estimation (Zero-DCE) algorithm [Nie et al., AAAI 2020] to enhance low light regions and contextual backgrounds in scenarios where the input video or frame suffers from poor lighting conditions. This method operates without any paired training data or reference images and learns enhancement curves in an unsupervised manner, making it highly suitable for real-world applications with varying illumination.



c)



Fig. 3. (a) Frames with lip synchronization produced by MuseTalk, (b) Raw input video clip without any processing, (c) Output from CodeFormer showing enhanced and restored faces from the MuseTalk frames.

Fig 3 presents a visual comparison across three stages of the video synthesis pipeline: the unprocessed input video, the lip-synced output generated by MuseTalk, and the enhanced frames refined using CodeFormer.

In the raw input frames (a), the subject’s face remains completely static, with no observable lip motion, reflecting the original video clip without any synchronization with audio.

The MuseTalk-generated frames (b) introduce audio-driven lip movements, successfully animating the subject to appear as if speaking. However, while the synchronization aligns with the speech input, the visual quality of these frames suffers from noticeable blurring, especially around the lips and teeth. This degradation affects the sharpness of the mouth region and slightly distorted facial detail, making the output less visually appealing and realistic.

The CodeFormer-enhanced frames (c) exhibit a clear improvement in both facial clarity and definition. The lips and teeth are more distinctly outlined, and the overall sharpness of the face is significantly improved. This enhancement corrects the blurriness introduced in the previous stage and contributes to a more polished and realistic appearance. Such refinement is particularly important in scenarios that demand high visual fidelity, such as virtual avatars, real-time communication, or video-driven animation systems. Now we will look at the quantitative performance of the Musetalk inference for the video.

To assess the feasibility of using MuseTalk in real-time streaming applications, we evaluated its performance under two different modes: one where each generated frame is saved to disk, and another where frames are processed and streamed directly without being written to storage. Both tests were conducted on an 8-second video clip synchronized with an 8-second audio track, rendered at 20 frames per second, totaling 160 frames.

When storing each frame to disk, the system experienced a notable slowdown due to frequent I/O operations, resulting in an average generation speed significantly below the real-time threshold. In contrast, the streaming mode, which bypassed disk writes and maintained all data in memory, allowed MuseTalk to operate much faster, comfortably exceeding the real-time frame rate.

This comparison highlights the impact of disk I/O as a bottleneck and emphasizes the efficiency gains achieved through a purely in-memory pipeline, making MuseTalk viable for applications such as live avatar communication or virtual meeting assistants. The tests were run on an Nvidia Tesla A40 GPU, available on the IIT Kanpur GPU server.

Mode	Average Output Speed (FPS)
With Saving to Disk	10.08
Real-Time (No Disk Saving)	29

To improve the quality of facial regions, especially under challenging lighting and motion conditions, we integrated **CodeFormer**, a robust face restoration framework that leverages a controllable latent code prediction mechanism to balance between quality and fidelity. This is particularly valuable in our pipeline for refining facial landmarks after low-light enhancement.

Inference Time & Performance:

- The model processed **201 frames** in **474.40 seconds**, resulting in an average inference speed of approximately **2.36 seconds per frame**.
- All experiments were run on a **Google Colab instance with an NVIDIA T4 GPU**.

IV CONCLUSION

In this work, we present a robust and real-time lip-syncing avatar pipeline tailored for online meetings, addressing key challenges of visual degradation and synchronization under poor lighting conditions. By integrating MuseTalk for expressive lip-sync generation, CodeFormer for artifact correction, and a custom preprocessing algorithm for illumination enhancement, our pipeline significantly improves the realism, consistency, and usability of avatar-based communication. The proposed system achieves superior performance in both subjective and objective evaluations, demonstrating its potential for seamless deployment in real-world virtual meeting applications.

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